

Geometry-Aware Conformal Decoding for Motor BCI Systems

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Introduction Brain-computer interface (BCI) motor decoders need accurate kinematic commands and reliable uncertainty. In closed loop, velocity uncertainty can gate or scale commands or give users feedback. Prior non-invasive BCI work used conformal prediction to defer uncertain discrete exoskeleton commands (Eliades and Papadopoulos, 2019); invasive work remains limited to one fixed-width global interval around 1D reach direction or position (Wei et al., 2024). We instead compare complete decoder-geometry systems, not geometries around a common predictor, for continuous 2D velocity decoding.

Task We used three sessions from each of two macaques (Indy, M1; Loco, M1+S1) in the Nonhuman Primate Reaching dataset (O’Doherty et al., 2017). Native 250-Hz kinematics were exposed every 12 ms, aligned with 12-ms causal neural windows (MLP: 252-ms context via 3 contiguous 84-ms channel-wise spike count windows; LSTM: trained recurrently on individual, variable-length reaches), for pseudo real-time 2D velocity decoding. Whole reaches were split chronologically, and each recording was analyzed separately. Across recordings, train/calibration/test ranges were 318–511/80–128/160–257 reaches and 27,119–64,348/8,653–18,469/17,352–44,248 time bins for Indy; 191–287/52–74/103–148 reaches and 25,989–40,510/13,287–31,735/25,492–45,848 time bins for Loco. Bin ranges span backbones. Each family used two-layer MLP and LSTM backbones (48 and 32 units/layer) with 0.5 dropout, trained over 10 seeds with AdamW (lr = weight decay = 10^{-3}) for up to 100 epochs (patience 15 on a validation set chronologically preceding calibration).

Probabilistic Decoders Cartesian CQR predicts 0.05/0.95 component quantiles; their midpoint is the command, and the ℓ_∞ norm of per-axis normalized residuals defines an axis-aligned rectangle (Romano et al., 2019). Gaussian-Mahalanobis outputs a mean and full covariance, Cholesky-parameterized (Muschinski et al., 2024); its Mahalanobis score yields an oriented ellipse (Messoudi et al., 2022). The polar likelihood factorizes velocity into truncated-normal non-negative speed and von Mises direction; its Cartesian expectation is the command, and the maximum of exact radial/angular negative log-likelihoods defines an annular sector. The geometries align uncertainty with V_x/V_y axes, principal covariance directions, and speed and direction, respectively (Fig. 1); a polar annulus can even span opposing directions while excluding zero speed, a property impossible for convex sets.

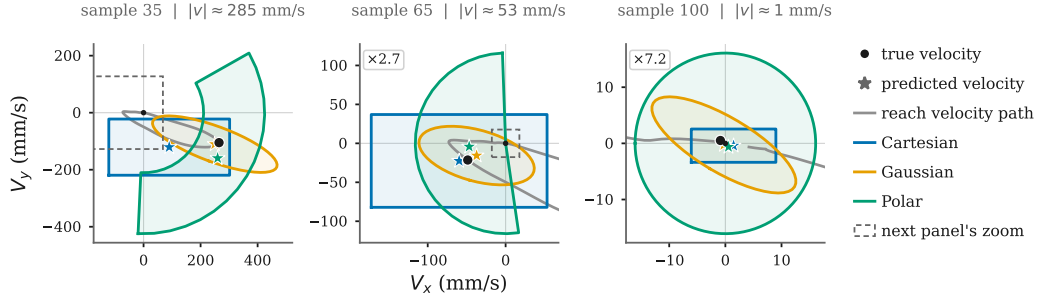


Figure 1: Symmetric regions for three samples shown at nested scales (Indy, LSTM, seed 73).

Calibration Joint coverage is $Y = (V_x, V_y) \in C(X)$, targeting 0.9 via split-conformal quantiles (Lei et al., 2018; Shafer and Vovk, 2008). Each family uses one global threshold or separate low/high-speed Mondrian thresholds (Boström and Johansson, 2020), split at 8 mm/s. Partitioning uses predicted speed; ground-truth speed strata are post-hoc diagnostics without formal guarantees. Per-component scores are coupled via Messoudi’s empirical copula (Messoudi et al., 2021) (Gaussian is inherently joint). Each 12-ms bin contributes a calibration score; scores are dependent: neural inputs and kinematics are autocorrelated, adjacent MLP inputs share 240 of 252 ms, and test-stream LSTM states persist without reset. Thus bin-level exchangeability is unjustified, making coverage empirical.

Results Global calibration was near nominal for Indy but undercovered Loco; low-speed coverage always exceeded 0.96, exposing the high-speed failure in Table 1. For Loco Mondrian narrowed this gap but enlarged regions. This contrast may reflect the recordings’ speed distributions: 52% of Indy versus 32% of Loco test bins exceeded the cutoff. Among high-speed samples, regions both covered truth and excluded zero at rates of 0.13–0.24/0.09–0.27 globally and 0.05–0.23/0.05–0.15 with Mondrian (Indy/Loco). Copula coupling had little effect on Cartesian regions. For Polar it reduced Indy area relative to both global and Mondrian, brought marginal coverage closer to nominal from above, and improved high-speed coverage; on Loco, it matched Mondrian’s high-speed coverage using smaller regions, but marginal coverage remained below nominal. These results show that calibration targets joint coverage, while geometry and correction allocation shape how regions could support closed-loop BCI; closed-loop performance gains remain untested in our open-loop evaluation.

Table 1: Recording-balanced means over seeds and backbones; R^2 is recipe-independent. High-speed coverage (true-speed strata) shows mean $\pm$ SD across the three recordings; area is in $10^3 (\text{mm/s})^2$.

	Calibration	Indy				Loco			
		R^2	Marg.	High-speed	Area	R^2	Marg.	High-speed	Area
Deterministic	–	0.65	–	–	–	0.57	–	–	–
Gaussian	Global	0.65	0.916	0.84 ± 0.02	15.0	0.57	0.882	0.64 ± 0.16	2.5
	Mondrian		0.917	0.84 ± 0.02	18.6		0.881	0.70 ± 0.12	4.4
Cartesian	Global	0.65	0.916	0.84 ± 0.02	19.3	0.54	0.889	0.70 ± 0.11	3.8
	Mondrian		0.917	0.85 ± 0.02	22.9		0.891	0.75 ± 0.09	5.3
	Copula		0.915	0.84 ± 0.02	19.1		0.888	0.70 ± 0.11	3.8
Polar	Global	0.64	0.911	0.84 ± 0.03	26.6	0.54	0.884	0.67 ± 0.13	4.1
	Mondrian		0.911	0.84 ± 0.01	34.5		0.891	0.76 ± 0.06	8.1
	Copula		0.904	0.87 ± 0.02	23.8		0.884	0.76 ± 0.09	5.2

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References

- Henrik Boström and Ulf Johansson. Mondrian conformal regressors. In Alexander Gammernan, Vladimir Vovk, Zhiyuan Luo, Evgueni Smirnov, and Giovanni Cherubin, editors, *Proceedings of the Ninth Symposium on Conformal and Probabilistic Prediction and Applications*, volume 128 of *PMLR*, pages 114–133. PMLR, 2020. URL <https://proceedings.mlr.press/v128/bostrom20a.html>.
- Charalambos Eliades and Harris Papadopoulos. Applying conformal prediction to control an exoskeleton. In Alex Gammernan, Vladimir Vovk, Zhiyuan Luo, and Evgueni Smirnov, editors, *Proceedings of the Eighth Symposium on Conformal and Probabilistic Prediction and Applications*, volume 105 of *PMLR*, pages 214–227. PMLR, 2019.
- Jing Lei, Max G’Sell, Alessandro Rinaldo, Ryan J. Tibshirani, and Larry Wasserman. Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523):1094–1111, 2018. doi: 10.1080/01621459.2017.1307116.
- Soundouss Messoudi, Sébastien Destercke, and Sylvain Rousseau. Copula-based conformal prediction for multi-target regression. *Pattern Recognition*, 120:108101, 2021. doi: 10.1016/j.patcog.2021.108101.
- Soundouss Messoudi, Sébastien Destercke, and Sylvain Rousseau. Ellipsoidal conformal inference for multi-target regression. In Ulf Johansson, Henrik Boström, Khuong An Nguyen, Zhiyuan Luo, and Lars Carlsson, editors, *Proceedings of the Eleventh Symposium on Conformal and Probabilistic Prediction with Applications*, volume 179 of *PMLR*, pages 294–306. PMLR, 2022. URL <https://proceedings.mlr.press/v179/messoudi22a.html>.
- Thomas Muschinski, Georg J. Mayr, Thorsten Simon, Nikolaus Umlauf, and Achim Zeileis. Cholesky-based multivariate gaussian regression. *Econometrics and Statistics*, 29:261–281, 2024. doi: 10.1016/j.ecosta.2022.03.001.
- Joseph E. O’Doherty, Mariana M. B. Cardoso, Joseph G. Makin, and Philip N. Sabes. Nonhuman primate reaching with multichannel sensorimotor cortex electrophysiology, May 2017. URL <https://doi.org/10.5281/zenodo.788569>. Dataset.
- Yaniv Romano, Evan Patterson, and Emmanuel J. Candès. Conformalized quantile regression. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019. arXiv:1905.03222.
- Glenn Shafer and Vladimir Vovk. A tutorial on conformal prediction. *Journal of Machine Learning Research*, 9:371–421, 2008.

Ganchao Wei, Zeinab Tajik Mansouri, Xiaojing Wang, and Ian H. Stevenson. Calibrating Bayesian decoders of neural spiking activity. *The Journal of Neuroscience*, 44(18): e2158232024, 2024. doi: 10.1523/JNEUROSCI.2158-23.2024.